

Marwan Ghazal

AI Student

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PROFILE

AI undergraduate building LLM-powered assistants and automations using Python, REST APIs, and structured JSON outputs. Experienced integrating OpenAI and Gemini APIs, designing prompt and tool workflows, and delivering end-to-end systems across data ingestion, processing, API development, and user-facing interfaces. Familiar with RAG and embeddings and currently building hands-on experience through applied projects. Fast, iterative builder using AI development tools such as Cursor, Windsurf, ChatGPT, and Claude.

SKILLS

Languages

Python, JavaScript (Node.js), SQL

RAG & Retrieval

chunking/summarization, embeddings concepts, vector search (FAISS concept), grounding & hallucination reduction

Databases & Platforms

PostgreSQL, MySQL, Supabase, Vercel

Dev Tools

Git/GitHub, Cursor, Copilot, ChatGPT, Windsurf

Backend & APIs

FastAPI, Flask, REST, JSON

LLMs / AI

OpenAI API, Gemini API, locally hosted Llama-family models via Hugging Face (Transformers), prompt engineering (system prompts, schema-based JSON outputs), cost-aware prompting.

Data/ML

Pandas, NumPy, scikit-learn, TensorFlow/Keras, XGBoost, Pytorch

PROFESSIONAL EXPERIENCE

AI Intern — National Training Academy (NTA)

08/2025 – 09/2025

- Delivered two production-ready AI systems: a **CV ranking pipeline** and an **OCR-based National ID extraction web app** for digital transformation workflows.
- Built an AI-powered CV ranking system using **Llama 3.1 + LoRA**, evaluating candidate-job fit via NLP analysis.
- Developed OCR pipelines using **OpenCV + Tesseract**, including preprocessing, correction, and extraction logic.

Impact: automated candidate shortlisting workflows and enabled reliable Arabic ID extraction via web APIs.

Junior Teaching Assistant — Nile University

02/2025 – 06/2025

- supported 50+ students with debugging and problem-solving guidance.
- Explained complex programming concepts clearly and efficiently, helping students resolve issues faster

PROJECTS

NeuroVista — EEG Session Analytics + LLM-Assisted Recommendations (Prototype)

Python, Claude API

- Built an end-to-end pipeline to ingest **EDF EEG recordings**, compute band-power trends (theta/alpha/SMR/beta), and generate session summaries.
- Implemented **rule-based scoring** layer to produce structured JSON per block/session (quality, stability, artifact heuristics).
- Integrated the **Claude API** to convert metrics into clinician-facing summaries and recommendations via schema-driven prompts with controlled outputs.
- Extending toward **RAG grounding** to reduce hallucinations and improve traceability.

CV Ranking Automation (LLM + LoRA Adapter)

Python, Flask

- Built a CV-to-job ranking service using **Llama 3.1 + LoRA**, generating structured JSON scores for automation.
- Designed an **NLP pipeline** to analyze and rank CVs against job descriptions.
- Added **multilingual** support

Career Guidance AI System

Python, Flask, LLM APIs

- Built a quiz-based system that calls **Gemini API** for career assessment and reasoning.
- Implemented CV analysis workflows using **ChatPDF API** with structured AI responses.
- Designed REST endpoints for submission, retrieval, and persistence; stored user profiles, answers, and recommendation history in **PostgreSQL**.

Egyptian National ID OCR

FastAPI, OpenCV, Tesseract

- Built a hybrid OCR pipeline with preprocessing (grayscale/blur/adaptive thresholding/perspective correction) and field-level validation.
- Delivered a simple web interface + API for real-time extraction.

Smart Food Guardian — Multimodal Nutri-Score (A–E)

Python, TensorFlow, XGBoost

- Developed a **multimodal machine learning system** combining tabular nutrition data, textual metadata, and product images.
- Demonstrated strong understanding of embeddings, modality fusion, and ML experimentation.
- Achieved 0.87 test accuracy / 0.83 macro-F1 on 5-class prediction.

EDUCATION

B.Sc. Artificial Intelligence — Nile University

Junior, Expected 2027

GPA: 4.0

EXTRACURRICULAR ACTIVITIES

Head of Public Relations (PR) | E-Club (Entrepreneurship Club), Nile University

- Led PR strategy and campaigns for club initiatives and flagship events, ensuring consistent messaging across channels.
- Managed sponsor/partner and guest-speaker communications (outreach, follow-ups, coordination).
- Trained PR team members on outreach messaging, communication etiquette, and content standards.

Volunteer – Wessal Club, Nile University

- Participated in charity initiatives by gathering clothes for donation.
- Co-led a Ramadan fundraising campaign and helped distribute **1,750** meals to people in need.

PR Committee Member – Book Club, Nile University

- Promoted club activities through targeted outreach and social content to boost engagement.